

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain Thevenin's & Norton's theorem in detail.

Q.24 Explain the working principle of

a) Nickel-Cadmium batteries

b) Silver-Oxide batteries

Q.25 Define r.m.s. value, maximum value, form factor and peak factor of AC Circuits.

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Roll No.

1st Year / Computer Engg.

Subject : Fundamental of Electrical & Electronics Engineering

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 The working principle of transformer is based on :

a) Mutual induction

b) Self Induction

c) Electromagnetic induction

d) Lenz's Law

Q.2 PET stands for

a) Field effect Transistor

b) Field effect Transistor

c) Field Emitter Transistor

d) None of these

Q.3 Which of the following is the electrolyte used in a lead-acid battery?

- a) Nitric Acid b) Hydrochloric Acid
- d) Sulphuric Acid d) Phosphoric Acid

Q.4 BJT stands for

- a) Bijunction Transistor
- b) Bipolar Junction Transistor
- c) Binary Junction Transistor
- d) None of these

Q.5 The frequency of AC is

- a) 0 Hz b) 50 Hz
- c) 100 Hz d) 120 Hz

Q.6 What are the two types of charge carriers in a semiconductor?

- a) Electrons and Protons
- b) Electrons & Neutrons
- c) Electrons & Holes
- d) Holes & Protons

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 CMOS stands for _____.

Q.8 What is Flux.

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Q.9 What is secondary cell?

Q.10 What is Ohm's Law?

Q.11 Draw the symbol of NPN Transistor.

Q.12 Define Inductance.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Differentiate between resistivity & conductivity.

Q.14 Explain power in pure resistance and inductance?

Q.15 Compare JEFT, MOSFET & BJT?

Q.16 Explain Kirchoff's Voltage Law.

Q.17 Explain the construction detail of single phase transformer.

Q.18 Explain the mechanism of current flow in PNP and NPN transistors with their symbols?

Q.19 What is Faraday's Law of electro-magnetic Induction?

Q.20 What are the principles of self & mutual induction?

Q.21 What are the applications of solar cells?

Q.22 Difference between AC and DC?

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